

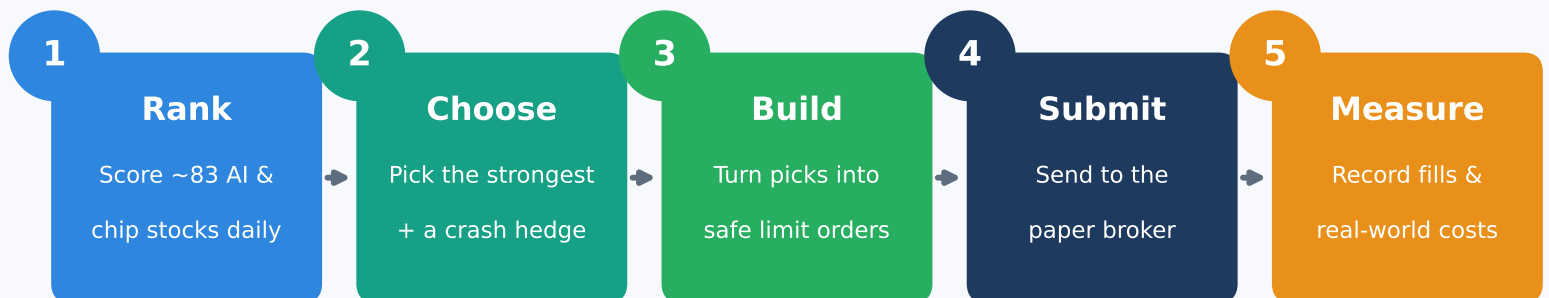
How the model_v4 system works

From a stock-picking model to simulated broker trades —
and what it would take to use real money

PAPER TRADING ONLY

fake money · real prices & timing · no capital at risk

The whole journey in five steps



In one sentence

Every day the model decides which trending tech stocks to hold (and when to hedge), those decisions become carefully-priced orders, and a simulated broker account fills them with real market prices — so we learn how the strategy would actually behave before a single real dollar is ever involved.

What model_v4 decides

~83 stocks

AI · semiconductors · tech

scored 0 to 1 every day on
momentum (who's trending up)



The strongest names

score above 0.70 = buy
above 0.90 = conviction add

Built-in risk brakes

Spread out

Money split evenly across
holdings — never more
than ~75% invested

Volatility governor

Automatically invests LESS
when markets get wild,
more when they calm

Stop-loss

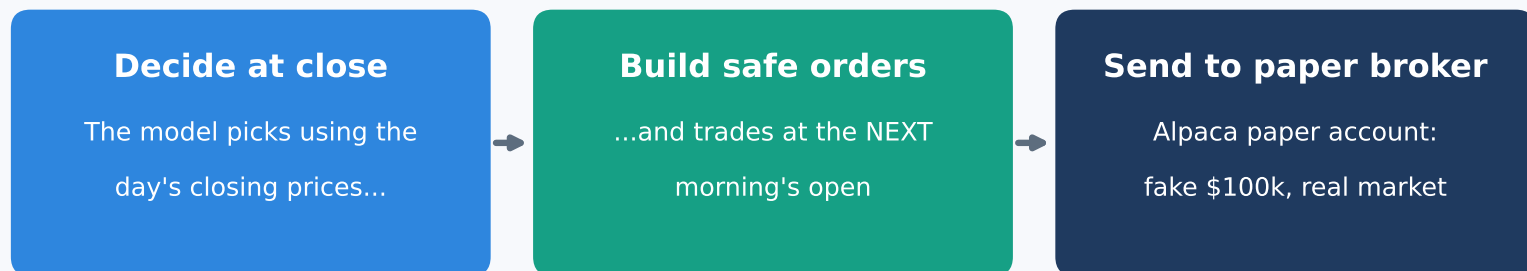
Exits a holding that falls
too far and is no longer
one of the leaders

The crash hedge (a shield, not a bet)

When the market jumps hard AND turbulence spikes at the same time — a classic "a pullback may be coming" warning — the system buys a ONE-DAY position (SOXS) that profits if semiconductors drop, cushioning the portfolio. It sits ON TOP of the model without changing how the model picks stocks.

Result: hold the trend when it's working, automatically pull back when it isn't.

Turning a decision into a (paper) trade



The key safety idea: a price 'collar'

Every order is a LIMIT order with a guardrail sized to how much each stock normally moves overnight — so we never overpay on a bad opening jump.

Calm stock

tight collar (~1%)

Jumpy hedge (SOXS)

wide collar (~6%)

If the open gaps past the collar, the order simply doesn't fill — we skip it.

Safety rails (always on)

- ✓ PAPER ONLY — the connection physically refuses any non-paper account
- ✓ SUBMIT IS OFF BY DEFAULT — sending real orders must be explicitly armed each time
- ✓ EVERY ACTION LOGGED — orders, fills, and costs are all recorded
- ✓ BROKER IS THE TRUTH — the ledger is rebuilt from what actually filled

What happens automatically, every weekday



What 'slippage' means (and why we measure it)

Slippage = the gap between the price we expected and the price we actually got. It's the hidden real-world cost of trading. The simulation assumes a small fixed cost; this pipeline measures the TRUE cost so we know if the edge survives it.

Two books, side by side

A frictionless simulation (assumes ~0.1% cost) vs. the live paper book (real fills & costs). Comparing them answers the only question that matters before **real money: does the strategy still win once trading frictions are real?**

A real run, end to end

1. Preview (dry-run), then submit for real

Top command = a dry-run: it prints the plan and sends NOTHING. Bottom = the same command with --live, which submits the 3 limit-on-open orders to the paper broker.

```
(.env) david@Davids-MacBook-Air ai-paper-trader % python src/broker_cron.py --phase open --account alpaca_tracker
2026-06-21T16:36:39 PHASE=open asof=2026-06-21 OK (dry-run) - 3 orders, 0 submitted
2026-06-21T16:36:39 BUY ARM qty 19 opg limit $455.17 collar 3.2% (tgt 19/now 0)
2026-06-21T16:36:39 BUY MRVL qty 27 opg limit $306.92 collar 3.5% (tgt 27/now 0)
2026-06-21T16:36:39 BUY SOXS qty 273 opg limit $3.80 collar 6.5% (tgt 273/now 0)

(.env) david@Davids-MacBook-Air ai-paper-trader % BROKER_ADAPTER_ALLOW_SUBMIT=yes python src/broker_cron.py --phase open --account alpaca_tracker --live --prequeue
2026-06-21T16:36:50 PHASE=open asof=2026-06-21 OK (LIVE) - 3 orders, 3 submitted
2026-06-21T16:36:50 BUY ARM qty 19 opg limit $455.17 collar 3.2% (tgt 19/now 0)
2026-06-21T16:36:50 BUY MRVL qty 27 opg limit $306.92 collar 3.5% (tgt 27/now 0)
2026-06-21T16:36:50 BUY SOXS qty 273 opg limit $3.80 collar 6.5% (tgt 273/now 0)
```

2. The orders land in the (paper) brokerage account

Alpaca shows all three orders ACCEPTED and queued for the next open (note the blue "no real money is being used" banner). These are the exact prices the model chose.

The screenshot shows the Alpaca Paper Trading interface. A blue banner at the top states: "You are on Paper Trading, no real money is being used." Below this, the "Orders" section is active, displaying a table of three orders. The table has columns for Asset, Order Type, Side, Qty, Filled Qty, Avg. Fill Price, Status, Source, Submitted At, Filled At, and Expires At. All three orders are in "accepted" status and are limit orders.

Asset	Order Type	Side	Qty	Filled Qty	Avg. Fill Price	Status	Source	Submitted At	Filled At	Expires At
SOXS	Limit @ \$3.80	buy	273.00	0.00	-	accepted	-	Jun 21, 2026, 04:36:50 PM	-	Jun 22, 2026, 03:00:00 PM
MRVL	Limit @ \$306.92	buy	27.00	0.00	-	accepted	-	Jun 21, 2026, 04:36:50 PM	-	Jun 22, 2026, 03:00:00 PM
ARM	Limit @ \$455.17	buy	19.00	0.00	-	accepted	-	Jun 21, 2026, 04:36:49 PM	-	Jun 22, 2026, 03:00:00 PM

What this proves

The full loop works on real market infrastructure — the model's picks became priced orders that a brokerage accepted — while every dollar stays simulated.

Real money would only follow after the checkpoints on the next page.

What it would take to use real money

PAPER

where we are

REAL

only if it earns it

cross the bridge only after every checkpoint below is met

1

Prove it over time

Run on paper for many months across calm AND turbulent markets — not just a lucky stretch.

2

Confirm costs don't kill it

Real slippage must stay small enough that the live book still beats the benchmarks.

3

Start tiny

Begin with an amount you could lose entirely; scale only after live results match paper.

4

Hard risk limits

Set a maximum loss, position caps, and an automatic 'stop everything' rule in advance.

5

Know the real-world rules

Taxes, account types, and brokerage regulations differ for real-money trading.

Important

This is a personal research tool, not financial advice. Simulated and past results do NOT guarantee future returns. Only risk money you can afford to lose, and consider speaking with a qualified financial professional first.